

Retrofitting a commercial RF induction generator into a computer-controlled, vacuum-integrated annealing system for reactive-metal grain growth

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High-temperature vacuum annealing near a metal’s melting point drives controlled grain growth but normally requires expensive turn-key vacuum induction furnaces. We describe an open, generator-agnostic modernization stack that converts a bare commercial RF induction generator into a computer-controlled, vacuum-integrated, pyrometer-feedback annealing system. The contribution is the reproducible control/sensing/vacuum layer — LabVIEW/DAQ analog power control, dual-wavelength optical temperature feedback, a high-vacuum quartz-tube chamber, and an in-house machined graphite crucible/susceptor — which transfers between induction generators provided the generator exposes a monotonic analog power-control input. The reproducible build is anchored on the lab’s current, USA-sourced generator (CEIA “Power Cube” 6kW, procured through East Coast Induction); a ~1970 LEPEL high-frequency furnace served as the originating prototype. We report the system design, construction and operating procedures, and a performance validation grounded in ~100 archived anneal logs: a fixed-geometry power–temperature calibration ($R^2 = 0.991$ over 1200–1400 °C), repeatable 1200 °C / 12 h Ni4N5 processing (1201.2 ± 1.3 °C across 8 runs), stable long-duration control up to 40 h at 1325 °C, and SEM/EBSD/optical microstructural characterization of annealed reactive-metal specimens cross-referenced to their logged thermal histories. Complete design files, a costed bill of materials, control software, and the full run-log and characterization archives are openly available.

Keywords: induction furnace; grain growth; vacuum annealing; LabVIEW; pyrometer; open-source hardware

Status: working draft generated from the repository’s design files, standard operating procedure, parts list, schematics, and run logs, retargeted from HardwareX to *Review of Scientific Instruments* (REVTeX 4.2, aip,rsi). **[TODO:]**-marked items flag values or figures that still need to be confirmed or added before submission; outstanding bill-of-materials corrections from co-author review are tracked in the pull request and in Appendix A.

I. INTRODUCTION

High-temperature annealing near a metal’s melting point drives controlled **grain growth**, which is central to studying microstructure–property relationships in metals such as nickel and iron^{1–3}. Doing this without oxidizing the sample requires both high temperature (~1400–1500 °C) and high vacuum (or inert atmosphere)^{4,5}. Commercial vacuum induction annealing furnaces that meet these requirements are expensive turn-key systems, which puts them out of reach for many academic labs.

This work takes the opposite approach: rather than building or buying a complete furnace, we **retrofit and modernize a bare RF induction generator** into a full annealing system by adding (1) computer power control, (2) closed-loop optical temperature control, (3) a high-vacuum quartz-tube chamber, and (4) a machined graphite crucible/susceptor. The emphasis of the contribution is the **modernization stack itself**, which is deliberately **generator-agnostic**: the same control software, vacuum chamber, pyrometer feedback, crucible, and work-coil geometry transfer from one

induction generator to another. The only requirement a reader’s generator must satisfy is an **analog power-control input with a monotonic command→power response** over the usable range (e.g. 4–20 mA, 0–5 V, or 0–5 mA); near-linearity is convenient but not required, because the pyrometer feedback and PID calibration absorb any nonlinearity. That portability is the reproducible result and is what makes the design reusable by other labs that already own, or can buy, such an induction generator.

A second feature that broadens the system’s applicability is the **machined graphite crucible/susceptor**: because graphite couples strongly to the RF field and heats the charge indirectly by conduction and radiation, the furnace is not limited to electrically conductive samples — it can also process poorly- or non-coupling materials such as YSZ ceramic. This makes the same build useful well beyond reactive-metal annealing.

The modernization was first prototyped on a ~1970 LEPEL high-frequency induction furnace owned by the BYU Mechanical Engineering department — at the time the only departmental machine able to reach the Ni/Fe melting range. A “remote control” capability was added in August 2019 to drive the generator’s power from a computer instead of the front-panel knob. The lab subsequently moved the same stack onto a current, USA-sourced solid-state generator (CEIA “Power Cube” 6kW via East Coast Induction), whose Power Controller C-V3 Plus accepts a standard analog (4–20 mA) power setpoint. An intermediate, separately-purchased import

generator (CYSI GP-15A) was evaluated but did not meet the lab’s control requirements — the required analog remote-control input was lost across vendor model substitutions — and was retired. The reproducible build documented here therefore targets the current USA generator, with the LEPEL retrofit retained as historical context.

Compared with commercial vacuum annealing furnaces, this approach has **substantially lower capital cost** (the generator + heating head + controller + chiller package was quoted at $\approx \$16.5\text{k}$ in 2021, and the retrofit components are itemized in Appendix A, versus $\$50\text{k}$ – $\$200\text{k}+$ for turn-key systems), is repairable and modifiable by the user, and is broadly reusable: any lab with an induction generator and an analog power input can reproduce the control, vacuum, and temperature-sensing additions described below. The system is aimed at:

- materials labs needing **oxidation-free grain-growth anneals** of reactive metals (Ni, Fe) near their melting points;
- labs that own a **bare induction generator** and want to add computer setpoint control, optical feedback, and vacuum integration without buying a turn-key furnace;
- groups processing **poorly- or non-coupling charges** (e.g. YSZ ceramic) that can be heated indirectly through the graphite susceptor; and
- teaching/maker contexts wanting a **repairable, modifiable** high-temperature vacuum furnace built from documented, open design files.

All design files, control software, run logs, and characterization data are openly available: the development repository is <https://github.com/vertical-cloud-lab/custom-induction-furnace> and an archival snapshot is deposited at Zenodo ([10.5281/zenodo.20878017](https://doi.org/10.5281/zenodo.20878017)), with hardware design files under the CERN-OHL-S v2 license and control code under the MIT license (proposed).

A. Scope and reproducibility boundary

To be explicit about what this paper does and does not reproduce: **this work does not reproduce the RF generator internals**. It reproduces the *modernization layer* built around a compatible generator — the computer power control, optical (pyrometer) feedback, high-vacuum quartz-tube chamber, machined graphite crucible/susceptor, work-coil geometry, and the mechanical support fixturing. A reader supplies (or already owns) a compatible generator and rebuilds the layer described here.

B. Prior art and comparison

Table I situates the design among the available alternatives.

II. SYSTEM DESIGN AND DESCRIPTION

A. Architecture and control signal flow

The system comprises five subsystems (see the schematics in `extracted-context/schematic-induction_furnace.md` and `extracted-context/schematic_support_stand.md`, and the labeled overview in Fig. 5):

1. **RF generator + work coil.** A solid-state induction generator (current build: CEIA Power Cube PW3-90/50) drives a water-cooled copper work coil. The coil carries both the oscillating current and cooling water. Coil geometry ($\sim 3\text{in}$ tall, 2.5in ID, ~ 6.5 turns) was tuned to the sample/crucible stack⁶ (see `extracted-context/order_corrections_coils.md`).
2. **Vacuum + gas.** A quartz tube forms the chamber and is connected through a KF40 flange and flexible bellows to an Edwards T-Station 85 turbo-molecular pump ($\sim 1 \times 10^{-6}$ – 10^{-8} Torr). A manual vent valve and an inert-gas regulator allow controlled venting/backfilling to suppress oxidation of reactive metals. The vacuum connection stack, in order, is: **quartz tube** $\rightarrow$ **KF40 flange + optical window** $\rightarrow$ **flexible bellows** $\rightarrow$ **vacuum gauge** $\rightarrow$ **Edwards T-Station 85** $\rightarrow$ **vent / inert-gas backfill branch**.
3. **Optical temperature sensing.** A dual-wavelength (ratio) pyrometer (the canonical unit is a LumaSense IMPAC ISR 6, 800 – 2500°C ; the IMPAC brand is LumaSense’s) views the sample through a KF40 optical window, providing a non-contact temperature signal that is robust to emissivity changes and to the RF field that would corrupt thermocouples^{7–9}.
4. **Control / DAQ + LabVIEW.** A LabVIEW DAQ analog output sets generator power; the pyrometer signal is read back for closed-loop ramp/soak control. The VIs implement manual control, automated ramping, PID tuning, and email alerts (see `code/` in the repository). In the CEIA build the DAQ outputs **0–5 V** to a **voltage-to-current loop conditioner** that converts the command to the controller’s **4–20 mA** power-setpoint input (the LEPEL prototype instead took a 0 – 5mA command directly). The closed-loop **signal flow** is therefore: **LabVIEW setpoint** $\rightarrow$ **DAQ analog output (0–5 V)** $\rightarrow$ **V** $\rightarrow$ **I**

TABLE I. Prior art and the limitation each class of alternative presents.

Class	Examples	Limitation this design addresses
Commercial vacuum induction annealing furnaces	turn-key systems (\$50k-\$200k+)	Cost; closed/non-modifiable; vendor lock-in
Open non-induction annealing systems	tube/box furnaces with resistive heating	Lower max temperature and slower ramp than induction; no RF-susceptor heating of non-coupling charges
Induction generators sold bare	CEIA, Ambrell, etc.	Ship without integrated vacuum, optical feedback, or computer setpoint control — exactly the layer added here

conditioner (4–20 mA) → generator power → sample/crucible heating → pyrometer optical reading → DAQ analog input (scaled to °C) → PID controller → updated analog command. The same path runs **open-loop** (direct command ramp) while the temperature is below the pyrometer's $\sim 700^\circ\text{C}$ detection floor; once the pyrometer reports a **sustained valid reading above that floor**, control hands over to **closed-loop** PID on the optical temperature¹⁰.

5. **Mechanical support stand + crucible.** A bolted support stand (8 short + 4 long sections, 12 corner braces, 4 floor mounts) positions the chamber, coil, pyrometer holder, and quartz disc. An **in-house machined graphite crucible** holds the sample and acts as the RF susceptor.

The LabVIEW control front panel (Fig. 1) exposes the ramp-up, PID-soak, and cool-down setpoint tables, the PID gains, the inert-gas flow control, and live power-command and pyrometer-temperature traces used to run and monitor an anneal.

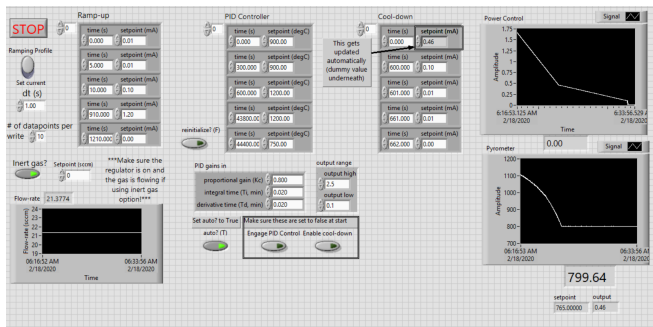


FIG. 1. LabVIEW control front panel (docs/equipment-reference/frontPanel.200219.PNG): editable ramp-up / PID-soak / cool-down setpoint tables (time vs. analog command in mA below the pyrometer floor and vs. °C once closed-loop), PID gains and output range, inert-gas flow setpoint/readback, and live power-command and pyrometer-temperature strip charts. Here the pyrometer reads $\sim 800^\circ\text{C}$ during ramp.

B. Technical specifications

Table II lists the technical specifications of the build documented here. These are *build-dependent* values: a reader supplies their own generator, pump, and pyrometer and re-establishes the corresponding values for their build.

C. Generator compatibility and portability

The control/sensing/vacuum layer transfers to any generator that meets these criteria:

- Exposes an **external analog power-control input** (e.g. 4–20 mA or 0–5 V/0–5 mA).
- The command→delivered-power response is **monotonic over the usable range** (near-linear is convenient but not required, since the pyrometer feedback closes the loop).
- Supports a **water-cooled induction head/coil** matching the documented coil dimensions.
- Delivers **enough RF power** to reach the target temperature for the documented crucible/sample stack.
- Allows **safe electrical isolation** between the DAQ/control path and the generator input.
- Has the required cooling and line power for the selected generator.

When porting to a new generator, the following must be re-established: analog-command scaling, power-to-temperature calibration, PID gains, coil/sample alignment, maximum safe ramp rate, and any generator-specific interlock behavior. Table III records which elements of the stack transferred unchanged from the ~ 1970 LEPEL prototype to the current CEIA build.

D. Custom-machined graphite crucible / susceptor

Because some charges of interest couple poorly to the RF field (e.g. YSZ ceramic) or must be

TABLE II. Technical specifications of the documented build. These are build-dependent values that a reader replaces with their own generator/pump/pyrometer equivalents.

Specification	Value (this build)
Induction generator (reproducible build)	CEIA “Power Cube” PW3-90/50 solid-state RF generator + Power Controller C-V3 Plus (East Coast Induction, USA)
Induction generator (prototype)	LEPEL high-frequency tube furnace (~ 1970), up to ~ 35 kW
Power-control input	4–20 mA analog setpoint via LabVIEW/DAQ $\rightarrow$ V $\rightarrow$ I conditioner (CEIA build; the LEPEL prototype used 0–5 mA — see Table III)
Generator-input requirement	External analog power-control input with monotonic command $\rightarrow$ power response over the usable range
Control signal chain	DAQ analog output (0–5 V) $\rightarrow$ voltage-to-current (4–20 mA) loop conditioner $\rightarrow$ generator power-setpoint input; pyrometer analog output $\rightarrow$ DAQ analog input (scaled to $^{\circ}\text{C}$)
Max sample temperature	~ 1400 – 1500 $^{\circ}\text{C}$ + (Ni/Fe melting range)
Pyrometer detection range	~ 800 – 2500 $^{\circ}\text{C}$ (dual-wavelength ratio); does not report below ~ 700 $^{\circ}\text{C}$
Demonstrated soak duration	minutes up to ~ 40 h (per <code>docs/data_log/</code> run logs)
Sample / crucible envelope	charge fits inside a 35 mm-ID quartz tube and a machined graphite crucible
Operating vacuum	$\sim 1 \times 10^{-6}$ to 1×10^{-8} Torr (turbo-molecular pump)
Temperature measurement	Dual-wavelength ratio pyrometer (LumaSense IMPAC ISR 6, ~ 800 – 2500 $^{\circ}\text{C}$)
Chamber	Quartz tube, KF40 optical-window flange, inert-gas vent
Crucible / susceptor	In-house machined graphite (3000 $^{\circ}\text{C}$ grade) crucible-susceptor
Work coil	Water-cooled copper, ~ 3 in tall, 2.5 in ID, ~ 6.5 turns

TABLE III. Transferability of the modernization stack: LEPEL prototype $\rightarrow$ CEIA reproducible build.

Aspect	LEPEL prototype (~ 1970)	CEIA reproducible build (current)	Transferred?
Analog control interface	0–5 mA	4–20 mA (Power Controller C-V3 Plus)	Concept yes; scaling re-set
Frequency	~ 200 – 450 kHz tube	solid-state (generator-set)	n/a (generator internal)
Max power	up to ~ 35 kW	6 kW	Re-tuned for charge
Pyrometer / optical feedback	dual-wavelength	same dual-wavelength stack	Yes
Vacuum chamber / quartz tube	quartz + KF40	same	Yes
Graphite crucible/susceptor	machined graphite	same	Yes
Support stand / fixturing	bolted stand	same	Yes
LabVIEW/DAQ control concept	manual + ramp/PID VIs	same VIs	Yes

physically contained, the sample is held in a **graphite crucible machined in-house from 3000 $^{\circ}\text{C}$ -grade graphite stock** (Cotronics 56L-3). The graphite couples strongly to the induction field and acts as a **susceptor**, transferring heat to the contained charge by conduction/radiation; this makes the same furnace usable for conductive metals (direct coupling) and for non/poorly-coupling materials (indirect, susceptor-driven heating). Crucibles are machined to fit inside the quartz tube and seat on a ceramic-cylinder / Teflon spacer stack. Cracked crucibles are repaired with a 3000 $^{\circ}\text{C}$ graphite cement (Resbond 931) rather than discarded. The two-piece geometry — a cup body and a lidded cap with a central bore — contains the charge while leaving an optical path for the pyrometer to sight the sample (Fig. 2). The measured dimensions of every piece in the stack (crucible body, lid, alumina

spacer discs, and sapphire pyrometer window) are given in Fig. 3; fully assembled, the crucible stack stands 13 mm tall. The specimen-loading sequence — specimen sandwiched between two alumina sheets in the cup body, lid seated on top, and sapphire window placed over the lid bore — is shown step by step in Fig. 4.

III. CONSTRUCTION AND OPERATION

A. Build sequence

1. **Support stand.** Assemble the bolted stand (8 short + 4 long sections, 12 corner braces, 4 floor mounts) and mount the generator/heating head and pyrometer holder.



FIG. 2. Fully disassembled graphite crucible. Top left is the crucible lid, top middle is the bottom of the crucible that acts as the specimen carrier, far right is the sapphire window, bottom left and middle are the alumina sample-surrounding sheets.

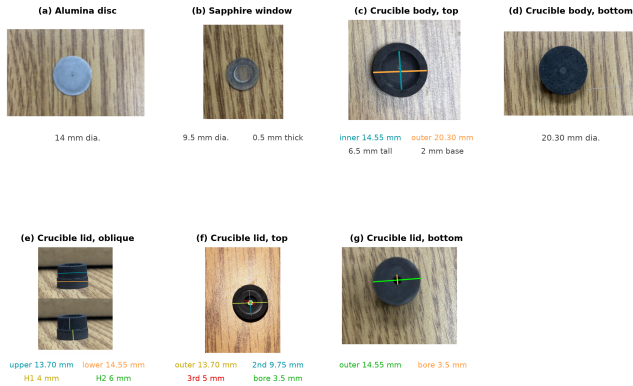


FIG. 3. Measured dimensions of each part of the graphite crucible/susceptor stack, with the coloured measurement lines drawn on the corresponding photograph. (a) Alumina spacer disc, 14 mm dia. (b) Sapphire pyrometer window, 9.5 mm dia., 0.5 mm thick. (c) Crucible body, top: machined sample cavity 14.55 mm ID (teal), 20.30 mm OD (orange); the cup is 6.5 mm tall with a 2 mm base. (d) Crucible body, bottom face, 20.30 mm dia. (e) Crucible lid, oblique: the stepped plug seats into the cavity — 13.70 mm upper step (teal), 14.55 mm lower shoulder (orange); the upper step is 4 mm tall (H1, yellow) and the lower shoulder 6 mm tall (H2, green). (f) Crucible lid, top: four concentric steps of 13.70 mm (outermost), 9.75 mm, 5 mm, and a 3.5 mm central pyrometer-sighting bore. (g) Crucible lid, bottom: 14.55 mm seating face (green) and the 3.5 mm through-bore (orange). Two alumina discs sandwich the specimen inside the cavity, the lid seats on top, and the sapphire window closes the bore so the pyrometer sights the sample through the lid; fully assembled the stack stands 13 mm tall. Measurements contributed by R. Guymon.

2. **Work coil.** Form/water-cool the copper coil to the corrected geometry (~ 3 in tall, 2.5 in ID, ~ 6.5 turns); connect to the generator head and the cooling-water circuit. Use the before/after figures from `induction_order_corrections.pptx` as the

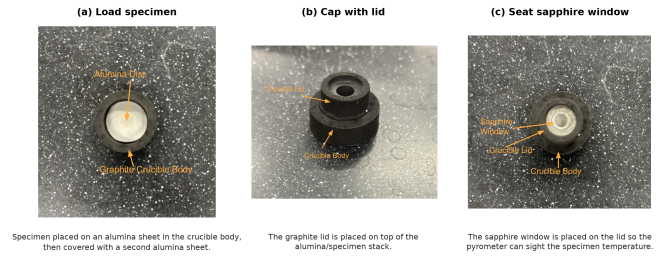


FIG. 4. Specimen-loading sequence for the graphite crucible/susceptor, in order (orange call-outs name the visible parts). (a) The specimen is placed on an alumina sheet in the crucible body and covered with a second alumina sheet, so the alumina discs isolate the specimen from the graphite. (b) The graphite lid is seated on top of the alumina/specimen stack. (c) The sapphire window is placed over the lid bore, closing the crucible while leaving an optical path for the pyrometer to sight the specimen temperature through the window. Photographs contributed by R. Guymon.

reference. [TODO: state the exact copper-tube OD/wall, bend radius/spacing method, and cooling-connection fittings from the coil drawing (`docs/coils-drawing.pdf`).]

3. **Graphite crucible/susceptor.** Machine the crucible from 3000 °C graphite stock to fit inside the quartz tube and accept the sample. Seat it on the ceramic-cylinder / Teflon spacer stack. Repair any cracks with graphite cement.
4. **Vacuum chamber.** Mount the quartz tube with the KF40 flange and optical window. The reference build seats a 55 mm quartz window on the overpressure KF40 centering ring and secures it with the plastic quick-clamp (reusable, no adhesive); a Torr-Seal'd blank flange is documented as an alternative for a permanent seal. Connect the flexible bellows to the Edwards T-Station 85 turbo pump; install the manual vent valve and inert-gas line.
5. **Control wiring.** Wire the DAQ analog output (0–5 V) through the voltage-to-current loop conditioner to the generator's analog power-control input (4–20 mA on the current CEIA controller; the LEPEL prototype took 0–5 mA directly). Wire the pyrometer output into the DAQ analog input and, for closed-loop ramping, the pyrometer/PLC temperature-control path (`docs/temp-control-modification/`). Maintain electrical isolation between the DAQ/control path and the generator interface.
6. **Software.** Deploy the LabVIEW VIs from `code/` in the repository and configure the DAQ channels; verify manual control before enabling automated ramp/soak.

[**TODO: curate a clean, photographed build sequence — candidate photos exist in docs/data-log/.../IFrun036_* and the PPTX media.**]

B. Operating procedure

Condensed from the lab SOP (extracted-context/SOP_induction_200901.md).

a. Preflight checklist (before energizing RF).

- ☐ Chamber at high vacuum ($\sim 1 \times 10^{-6}$ Torr or better) or backfilled to the intended inert atmosphere.
- ☐ Cooling-water flow confirmed at the coil/generator (no leaks).
- ☐ Coil/sample alignment and chamber clearance verified.
- ☐ Pyrometer aligned and sighted on the sample/crucible.
- ☐ No conductive objects (jewelry, watches, phones, tools) within ≥ 6 –12 in of the coil.

b. Startup.

1. **Load the sample.** With the chamber vented to atmosphere (never open the vent valve while the turbo pump is spinning), stack sample $\rightarrow$ optional alumina plate $\rightarrow$ ceramic cylinder $\rightarrow$ Teflon spacer inside the graphite crucible, insert into the quartz tube, and clamp the KF40 flange finger-tight with the O-ring.
2. **Pump down.** Close the vent valve, start the roughing/turbo pumps, and wait for high vacuum ($\sim 1 \times 10^{-6}$ Torr or better).
3. **Cooling water.** Check for leaks and confirm flow before applying RF power.

c. Normal run.

4. **Power on.** Bring up the generator per its sequence, then run the LabVIEW ramp/soak profile. Below the pyrometer's $\sim 700^\circ\text{C}$ detection floor the VI ramps **open-loop** on the analog command; once the pyrometer returns a **sustained valid reading above the $\sim 700^\circ\text{C}$ floor** (the handoff criterion), control switches to **closed-loop** PID on the optical temperature.
5. **Anneal.** Hold the temperature/time profile (e.g. $\sim 1200^\circ\text{C}$ for several hours for Ni grain growth); the VI can email status alerts.

d. Shutdown.

6. **Ramp down.** Ramp power down and stop RF, then turn off cooling water, allow the sample to cool under vacuum, and vent only after the turbo pump has fully stopped.

e. Abort / emergency stop.

- **Loss of cooling water or chiller alarm:** immediately command power to zero / stop RF, then investigate; do not re-energize until flow is restored.
- **Vacuum loss or quartz-tube fracture:** stop RF immediately; do not vent against a spinning turbo pump.
- **Arcing, smoke, or runaway temperature:** zero the analog command, kill RF at the generator, and lock out before inspecting.

C. Safety

The system combines high voltage/RF, high vacuum, high temperature, and water cooling; the principal hazards and mitigations are summarized in Table IV.

PPE / shielding. Operators should wear safety glasses (quartz-fracture/implosion risk), clean nitrile gloves when handling vacuum components, and remove all conductive personal items before approaching the energized coil; keep sensitive instrumentation outside the RF near-field.

IV. PERFORMANCE VALIDATION

The repository contains 100+ logged runs (docs/data_log/, IFrun001–100) across Ni200, Ni4N5, YSZ, and Pd thermal-evaporation experiments, with temperatures of ~ 900 – 1400°C and soak durations from minutes to ~ 40 h, each with .xlsx/.txt/.lvm traces and some photos. Per-run traces were parsed into uniform CSV/PNG files (docs/data_log/processed/build_run_traces.py), and the quantitative metrics below were computed reproducibly from those traces by paper/build_validation_figures.py (values mirrored in paper/validation-metrics.json). Because the archive is a retrospective operating record rather than a prospectively balanced design of experiments, every comparison is restricted to an internally consistent subset (matching material, nominal profile, and operating configuration) and the power–temperature relation is reported as an *operational* calibration for one fixed geometry, not as a universal transfer function.

a. *System overview (Fig. 5).* Figure 5 is a labeled schematic of the full system — recirculating water chiller, induction generator, heating head, water-cooled work coil, quartz-tube vacuum chamber with optical quartz disc, pyrometer holder/pyrometer line-of-sight, pressure sensor, bellows, and the Edwards T-Station 85 turbo-molecular pump on the bolted support stand, with the DAQ/control laptop closing the loop. (Hardware photographs are collected in paper/extracted-context/figures/; a labeled

TABLE IV. Principal hazards and mitigations.

Hazard	Failure mode	Consequence	Mitigation
High voltage / RF field	Contact with coil or induced currents in nearby conductors	Electric shock, burns	Keep conductive objects/electronics ≥ 6 –12 in from the coil while powered; isolate DAQ/control path
Lockout-tagout	Internal transformer-tap change (LEPEL prototype)	Shock during service	Follow LOTO before opening the generator
Vacuum / quartz	Venting against a spinning turbo pump; quartz fracture/implosion	Pump destruction; flying glass	Never vent while the turbo spins; handle quartz/gauge carefully; use eye protection
Water + electricity	Coil/chiller leak near energized hardware	Shock, equipment damage	Confirm flow and check for leaks before RF; route water away from electronics
Thermal	Sample melt-through; hot graphite oxidizing on vent	Pump-line contamination, burns	Cool under vacuum; let cool before venting; nitrile gloves when loading/unloading
Inert gas	Backfill/venting in an enclosed space	Asphyxiation	Adequate ventilation; do not backfill in confined areas
EMI	RF coupling into nearby instruments	Data corruption, instrument damage	Shield/separate sensitive electronics; keep them clear of the coil

composite photograph will replace this schematic on final submission.)

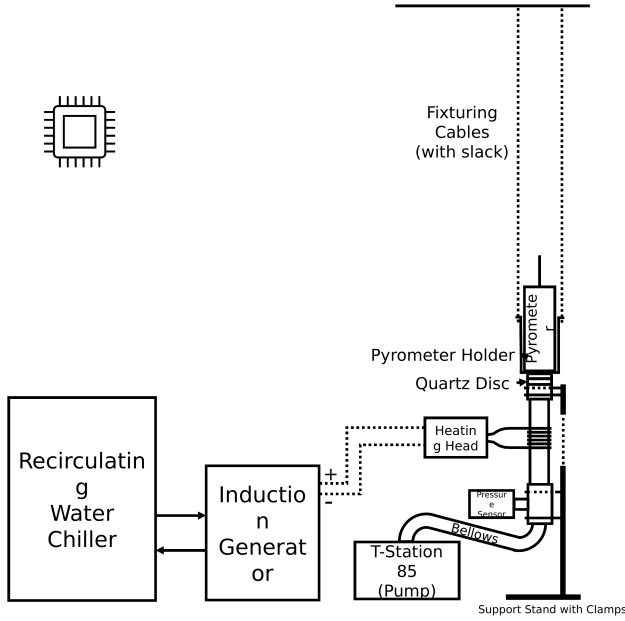


FIG. 5. System overview of the computer-controlled, vacuum-integrated induction annealing system: recirculating water chiller and induction generator drive a water-cooled heating head/work coil around a quartz-tube vacuum chamber; a dual-wavelength pyrometer sights the charge through a quartz optical disc; an Edwards T-Station 85 turbo-molecular pump (with pressure gauge and bellows) evacuates the chamber; and a DAQ/laptop sets generator power and reads the pyrometer for closed-loop control. Rendered from the editable source schematic `docs/induction-furnace-schematic.pptx`.

b. Power-command $\rightarrow$ temperature calibration (Fig. 6). For one fixed Ni4N5/quartz/graphite/optical

configuration, four archived steady-state anneals spanning 1200–1400 °C (IFrun039, IFrun040, IFrun038, IFrun032) give a monotonic relation between the soak-mean analog power command and the soak-mean pyrometer temperature. A linear guide-to-the-eye fit over these points is $T \approx 931 + 632I$ (°C, I in mA), $R^2 = 0.991$, $n = 4$. This is presented as an operational calibration only: the broader archive shows configuration-dependent offsets (e.g. runs near 1200 °C required soak-mean commands of ~ 0.44 mA vs. ~ 0.66 mA in a different era/geometry), so the fit must not be read as a universal command–temperature law.

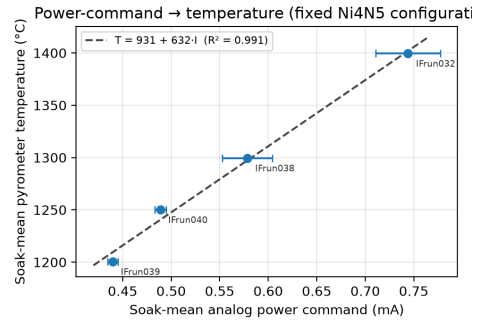


FIG. 6. Power-command $\rightarrow$ temperature calibration for one fixed archived Ni4N5 configuration. Points are soak-mean values; horizontal bars are the in-soak SD of the analog command. The dashed line is a guide-to-the-eye linear fit ($R^2 = 0.991$, $n = 4$).

c. Representative ramp/soak with control metrics (Fig. 7). Run IFrun079_Ni4N5.084.1300C.12h is a representative closed-loop anneal. Measured against the 1300 °C setpoint, the soak held 1302.1 ± 3.0 °C (mean $\pm$ SD) over the 12 h plateau, with a 10–90 % rise time of ~ 1135 s, an overshoot of ~ 12 °C, a maximum in-

soak deviation of $\sim 10^\circ\text{C}$, and a near-zero linear drift of -0.02°C/h , demonstrating that the pyrometer-feedback loop holds a programmed soak over many hours.

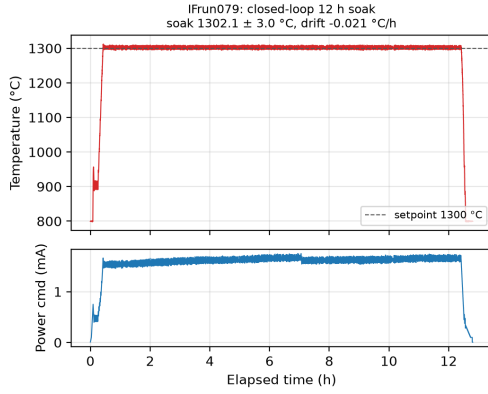


FIG. 7. Representative closed-loop anneal (IFrun079, Ni4N5, nominal 1300°C / 12 h): pyrometer temperature with setpoint overlay (top) and analog power command (bottom). Soak mean $1302.1 \pm 3.0^\circ\text{C}$, drift -0.02°C/h .

d. Repeatability and long-soak stability (Figs. 8–9). Run-to-run repeatability was assessed on the most internally consistent cohort available: eight Ni4N5 anneals at a nominal 1200°C / 12 h condition (IFrun052–IFrun060, excluding missing run numbers), all logged under the same gas-flow configuration. Across these runs the soak temperature was $1201.2 \pm 1.3^\circ\text{C}$ (range 1199.8 – 1204.2°C ; coefficient of variation 0.11%; $n = 8$), peak temperature $1212.6 \pm 6.7^\circ\text{C}$, and soak duration 766 ± 15 min. Runs from other periods or geometries (e.g. IFrun016, IFrun019, IFrun049) are deliberately *not* pooled into this estimate. Long-duration stability was demonstrated with two Ni200 anneals at 1325°C : a 20 h run (IFrun081) held $1326.5 \pm 1.5^\circ\text{C}$ with drift -0.004°C/h , and a 40 h run (IFrun080) maintained ~ 1326 – 1329°C over 2400 min in-band.

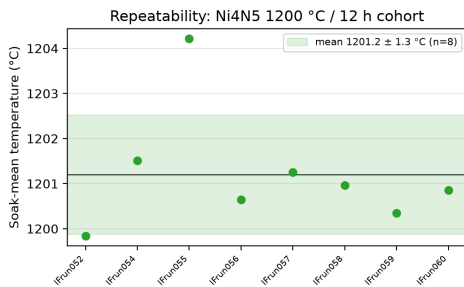


FIG. 8. Run-to-run repeatability of the Ni4N5 1200°C / 12 h cohort (IFrun052–IFrun060, $n = 8$). Points are per-run soak-mean temperatures; the band is mean $\pm$ SD ($1201.2 \pm 1.3^\circ\text{C}$).

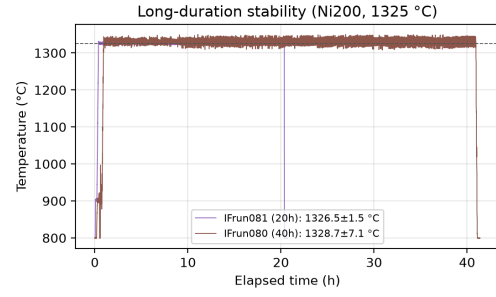


FIG. 9. Long-duration thermal stability: Ni200 anneals at 1325°C held for 20 h (IFrun081) and 40 h (IFrun080).

V. MICROSTRUCTURAL VALIDATION

The furnace exists to change microstructure, so the key validation is **real microstructural data** tied to specific thermal histories. The committed SEM/EBSD and optical-microscopy archives (docs/SEM/, docs/optical/) are keyed by Ni4N5_### / Ni200_### specimen IDs that map directly to the run IDs in docs/data_log/. This mapping is built reproducibly by paper/build.characterization-crossref.py, which parses the specimen ID out of every characterization folder, joins it to the machine-parsed run summary, and writes the per-specimen cross-reference in paper/characterization-crossref.csv/.md: **35 characterized specimens**, of which **9 link to a parsed run log**, **8 carry committed EBSD/OIM data**, and one carries EDS. Inverse-pole-figure (IPF) EBSD orientation maps of two annealed specimens (Fig. 10) resolve hundreds of indexed grains, and the best-correlated specimen, Ni4N5_081 (annealed in IFrun082 at a nominal 1300°C for 20 h; logged soak $\sim 1301^\circ\text{C}$), is documented across length scales — optical grain map with in-image grain-size measurements, SEM survey, and high-magnification grain-boundary detail (Fig. 11). The full multi-gigabyte raster/EBSD archive (including .oim, .osc, .ang, and DREAM.3D volumes) remains on the lab’s Box share and is inventoried per file in docs/SEM/CATALOG.csv and docs/optical/CATALOG.csv. Grain size is quantified with: number of grains analyzed, equivalent-circle-diameter or intercept metric, median + IQR (or mean $\pm$ SD), segmentation/cleanup criteria, minimum-grain-area threshold, and whether twins are merged or separated, with the specimen/run (not the individual grain) treated as the experimental unit^{1,3}. Where as-received baselines are unavailable for a batch, results are framed as final-state comparisons across archived anneals rather than as fully parameterized grain-growth kinetics.

Controls that further isolate the instrument’s behavior — open-loop vs. closed-loop on the same target cycle, vacuum vs. inert-gas backfill mode, directly-coupled metal vs. susceptor-mediated ceramic heating, and multi-hour long-soak stability — are partially represented in

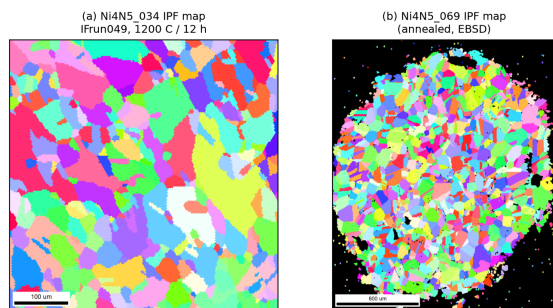


FIG. 10. EBSD inverse-pole-figure (IPF) orientation maps of two annealed Ni4N5 specimens, rendered from the committed docs/SEM/ archive. (a) Ni4N5_034 (IFrun049, 1200 °C / 12 h); (b) Ni4N5_069. Each indexed color is a crystallographic orientation; the maps resolve hundreds of grains for grain-size and texture analysis.

the archive and are natural extensions for a prospective study.

a. Specimen ↔ thermal-history linkage. Each microstructure specimen is tied to its exact run via Table V. “Nominal soak” is the target parsed from the run filename; “logged soak T” is the machine-parsed soak-mean from the trace (run_summary.csv); “atmosphere” indicates whether active gas flow was present in the parsed log summary (has_flow), not a fully reconstructed gas chemistry.

b. Characterization ↔ run inventory. Table VI lists the specimens that carry committed SEM/EBSD/optical data and a matched parsed run log (generated by paper/build_characterization_crossref.py). Two specimens (Ni4N5_034, Ni4N5_081) carry committed EBSD orientation maps on top of SEM/optical imaging, and Ni4N5_081 is the most completely characterized run (Fig. 11). The complete 35-specimen cross-reference, including characterized specimens whose run logs are not in the parsed set, is in paper/characterization_crossref.csv.

A. Limitations of the retrospective validation

The validation dataset is retrospective rather than prospectively designed, so the archived runs are not treated as a balanced factorial experiment and comparisons are restricted to internally consistent subsets. Because induction coupling, radiative view factors, and pyrometer line-of-sight depend on specimen/crucible geometry and optical alignment, the power-command/temperature calibration is reported only for a fixed archived configuration and is not a universal transfer function. Reported temperatures are closed-loop ratio-pyrometer readings in a fixed optical geometry; absolute specimen temperature may still depend on emissivity and surface condition not fully reconstructable for every archived run^{7,8}. Run-to-

run repeatability is evaluated only within the most consistent cohort (IFrun052–IFrun060); most other temperature/time conditions are singletons, so inference is limited to descriptive summaries. Microstructural comparisons are reported only for specimens whose image files link unambiguously to a logged thermal history; ambiguously labeled images are excluded rather than inferred.

VI. CONCLUSIONS

A bare commercial RF induction generator can be converted into a computer-controlled, vacuum-integrated, pyrometer-feedback annealing system with a documented, generator-agnostic modernization layer: LabVIEW/DAQ analog power control, dual-wavelength optical temperature feedback, a high-vacuum quartz-tube chamber, and an in-house machined graphite crucible/susceptor. Ported from a ~1970 LEPEL prototype to a current CEIA solid-state generator, the same stack demonstrated an operational power-temperature calibration ($R^2 = 0.991$ over 1200–1400 °C in a fixed geometry), run-to-run soak repeatability of 1201.2 ± 1.3 °C across eight nominally identical Ni4N5 anneals, and stable closed-loop soaks up to 40 h at 1325 °C, with annealed-specimen microstructures (SEM/EBSD/optical) cross-referenced to their logged thermal histories. Because the layer requires only a monotonic analog power-control input from the generator, the design is directly reusable by other laboratories at a small fraction of the cost of turn-key vacuum induction furnaces; complete design files, the costed bill of materials (Appendix A), control software, and the full run-log and characterization archives are openly available.

SUPPLEMENTARY MATERIAL

The development repository (<https://github.com/vertical-cloud-lab/custom-induction-furnace>; archival snapshot [10.5281/zenodo.2087801](https://zenodo.org/record/2087801)) contains the complete supplementary material: LabVIEW control VIs and MATLAB utilities (code/), system schematics and the coil drawing (docs/), the standard operating procedure, the full parts list, the 100+ raw and machine-parsed anneal logs (docs/data_log/), the SEM/EBSD and optical characterization archives with per-file catalogs (docs/SEM/, docs/optical/), and the scripts that reproduce every figure and metric in this paper (paper/).

ACKNOWLEDGMENTS

BYU Mechanical Engineering Department; the Johnson group; Kevin Cole. [TODO: funding

Ni4N5_081 microstructure after IFrun082 (1300 °C / 20 h, flow)

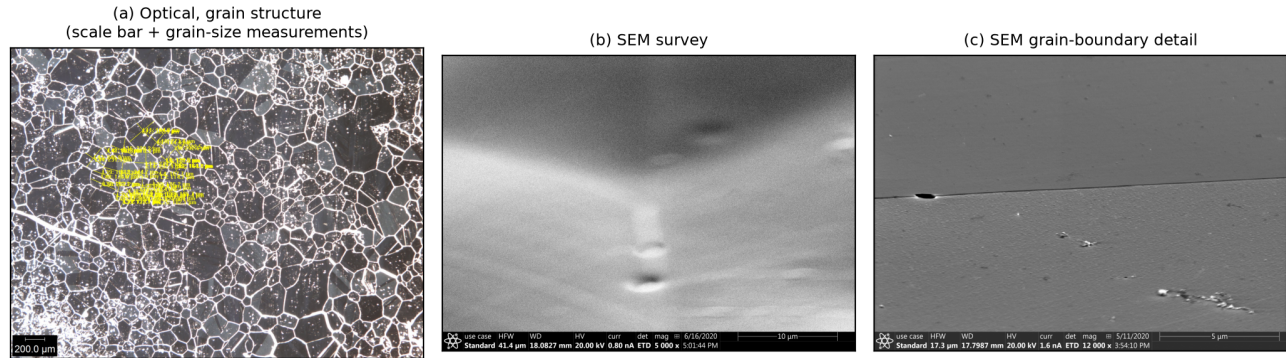


FIG. 11. Multi-scale microstructure of specimen Ni4N5_081 after IFrun082 (nominal 1300 °C / 20 h, gas flow), correlating the characterization archive with a single logged thermal history. (a) Optical micrograph (grain structure, 200 μm scale bar, with in-image grain-size measurements); (b) SEM survey; (c) high-magnification SEM grain-boundary detail. All panels are committed files under docs/optical/ and docs/SEM/.

TABLE V. Specimen ↔ thermal-history linkage for the runs used in the validation figures.

Specimen	Material	Run ID	Nominal soak	Logged soak T	Soak time	Atmosphere	Role / figure
Ni4N5_026	Ni4N5	IFrun039_Ni4N5_026_1200C_6h	1200 °C	1200.4 °C	6 h	no-flow	Calibration (Fig. 6)
Ni4N5_027	Ni4N5	IFrun040_Ni4N5_027_1250C_6h	1250 °C	1250.0 °C	6 h	no-flow	Calibration
Ni4N5_025	Ni4N5	IFrun038_Ni4N5_025_1300C_1h	1300 °C	1299.9 °C	1 h	no-flow	Calibration
Ni4N5_022	Ni4N5	IFrun032_Ni4N5_022_1400C_-10min	1400 °C	1399.6 °C	10 min	no-flow	Calibration
Ni4N5_084	Ni4N5	IFrun079_Ni4N5_084_1300C_12h	1300 °C	1302.1 °C	12 h	flow	Representative (Fig. 7)
Ni4N5_040	Ni4N5	IFrun052_Ni4N5_040_1200C_12h	1200 °C	1199.8 °C	12 h	flow	Repeatability (Fig. 8)
Ni4N5_042,044	Ni4N5	IFrun054_Ni4N5_042,044_-1200C_12h	1200 °C	1201.5 °C	12 h	flow	Repeatability
Ni4N5_045,046	Ni4N5	IFrun055_Ni4N5_045,046_-1200C_12h	1200 °C	1204.2 °C	12 h	flow	Repeatability
Ni4N5_047,048	Ni4N5	IFrun056_Ni4N5_047,048_-1200C_12h	1200 °C	1200.6 °C	12 h	flow	Repeatability
Ni4N5_049,050	Ni4N5	IFrun057_Ni4N5_049,050_-1200C_12h	1200 °C	1201.2 °C	12 h	flow	Repeatability
Ni4N5_051,052	Ni4N5	IFrun058_Ni4N5_051,052_-1200C_12h	1200 °C	1201.0 °C	12 h	flow	Repeatability
Ni4N5_053,054	Ni4N5	IFrun059_Ni4N5_053,054_-1200C_12h	1200 °C	1200.3 °C	12 h	flow	Repeatability
Ni4N5_056,057	Ni4N5	IFrun060_Ni4N5_056,057_-1200C_12h	1200 °C	1200.9 °C	12 h	flow	Repeatability
Ni200_015	Ni200	IFrun081_Ni200_015_1325C_20h	1325 °C	1326.5 °C	20 h	flow	Long soak (Fig. 9)
Ni200_017	Ni200	IFrun080_Ni200_017_1325C_40h	1325 °C	1326.3 °C	40 h	flow	Long soak
Ni4N5_079,080	Ni4N5	IFrun072_Ni4N5_079,080_-1400C_12h	1400 °C	—	12 h	flow	Process-window specimen

TABLE VI. Specimens with committed characterization data and a matched, parsed furnace run. “SEM”/“Opt.” are committed file counts; “EBSD” flags committed inverse-pole-figure/OIM data. Full inventory in paper/characterization-crossref.csv.

Specimen	Material	Matched run	SEM	Opt.	EBSD
Ni200_001	Ni200	IFrun004/005	0	3	no
Ni200_015	Ni200	IFrun081 (1325 °C/20 h)	0	3	no
Ni200_017	Ni200	IFrun080 (1325 °C/40 h)	0	2	no
Ni4N5_001b	Ni4N5	IFrun006	0	2	no
Ni4N5_010	Ni4N5	IFrun016 (1200 °C)	0	4	no
Ni4N5_012	Ni4N5	IFrun019 (1200 °C)	0	1	no
Ni4N5_034	Ni4N5	IFrun049 (1200 °C/12 h)	4	0	yes
Ni4N5_053	Ni4N5	IFrun059 (1200 °C/12 h)	3	3	no
Ni4N5_081	Ni4N5	IFrun082 (1300 °C/20 h)	10	10	yes

sources.]

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose. [TODO: confirm with all authors.]

Ethics Approval

This work did not involve human participants, human data, or animal subjects; ethics approval is not applicable.

Author Contributions

[**TODO: confirm CRediT roles**] — provisional: **Sterling G. Baird**: Conceptualization, Software, Investigation, Writing – original draft. **Kevin Cole**: Methodology, Resources, Supervision. Additional authors to be confirmed.

DATA AVAILABILITY

The data that support the findings of this study are openly available in the development repository `vertical-cloud-lab/custom-induction-furnace` (<https://github.com/vertical-cloud-lab/custom-induction-furnace>); an archival snapshot (design files, run logs, characterization data, and extracted context) is deposited at Zenodo, <https://doi.org/10.5281/zenodo.20878017>.

Appendix A: Bill of materials

Table VII itemizes the build. Groupings are denoted by the designator prefix: **GEN** = induction generator + controller (the current USA reproducible build; the legacy LEPEL is the prototype alternative, not a line item), **RETRO** = the reproducible retrofit (control / vacuum / sensing) parts, and **CONS** = consumables. Full vendor/price detail for the RETRO and CONS parts is in `extracted-context/parts_list.md` (extracted from `docs/induction_parts_list.xlsx`).

Costing conventions: all costs are in **USD**; prices are the lab’s purchase records (quote year ~2019–2021), several from used/surplus/eBay sources as noted; shipping and tax are excluded unless stated. **The build instructions and the bill of materials are canonical for the CEIA system only**; the legacy LEPEL prototype and the rejected CYSI import path are documented in the text and in Table III, not here.

[**TODO: confirm RETRO-1/4/5 pricing, the exact NI DAQ model, and the V→I loop-conditioner model, folding in the co-author**

review corrections (NI USB-6000 \$281; NI-9265 \$856 + NI-9203 \$1,181 + cDAQ-9174 \$1,718; Edwards nEXT T-Station 85H \$13,573.65; wide-range gauge D14701000 \$1,543.29; TAV5 \$652.80 new; pyrometer \$385 used). GEN line items are filled from East Coast Induction quote 210203AP (3 Feb 2021; generator + heating head + controller + chiller + line transformer total \$16,487).]

Appendix B: Design-file inventory

Table VIII inventories the open design files in the repository (paths relative to the repository root).

[**TODO: export the LabVIEW .vi block diagrams to PDF/PNG so non-LabVIEW readers can inspect the control logic.**]

- ¹Y.-X. Liu, Z.-J. Ke, R.-H. Li, J.-Q. Song, and J.-J. Ruan, “Study of grain growth in a ni-based superalloy by experiments and cellular automaton model,” *Materials* **14**, 6922 (2021).
- ²H. Mohrbacher, “Application of microalloying for controlling recrystallization and grain growth during downstream steel processing,” *Metallurgical and Materials Transactions A* **57**, 2548–2571 (2026).
- ³K. Alogab, D. Matlock, J. Speer, and H. Kleebe, “The influence of niobium microalloying on austenite grain coarsening behavior of ti-modified sae 8620 steel,” *Isij International* **47**, 307–316 (2007).
- ⁴A. F. Zhornyak and V. E. Olikier, “Reduction and decarburization during the annealing of atomized iron-carbon alloy powders,” *Soviet Powder Metallurgy and Metal Ceramics* **21**, 84–93 (1982).
- ⁵Y. Muramatsu, S. Wanikawa, M. Ohtaguchi, H. Okada, and F. Abe, “Gas contamination due to milling atmospheres of mechanical alloying and its effect on impact strength,” *Materials Transactions* **46**, 681–686 (2005).
- ⁶W. Vankan, N. van Hoorn, A. de Wit, and T. Koenis, “Electromagnetic induction heating of TP-CFRP laminates: FEM modelling and validation,” in *Proceedings of the 34th Congress of the International Council of the Aeronautical Sciences (ICAS)* (The International Council of the Aeronautical Sciences (ICAS), 2024).
- ⁷S. H. Ham, M. Ferte, G. Fricout, L. Depalo, and C. Carteret, “In-situ spectral emissivity measurement of alloy steels during annealing in controlled atmosphere,” *Quantitative InfraRed Thermography* (2016), 10.21611/qirt.2016.041.
- ⁸F. Suleiman, “Improving pyrometry of advanced high strength steels during intercritical annealing,” Unknown journal (2025).
- ⁹R. Usamentiaga, P. Venegas, J. Guerediaga, L. Vega, J. Molleda, and F. Bulnes, “Infrared thermography for temperature measurement and non-destructive testing,” *Sensors* **14**, 12305–12348 (2014).
- ¹⁰J. Meng, H. Gao, M. Ruan, H. Guo, X. Zhou, and D. Zhang, “Design of vacuum annealing furnace temperature control system based on ga-fuzzy-pid algorithm,” *PLOS ONE* **18**, e0293823 (2023).

TABLE VII. Bill of materials for the reproducible CEIA-anchored build (USD; see costing conventions in the text).

Desig.	Component	No.	Unit	\$	Total	Source	Material
GEN-1	CEIA “Power Cube” PW3-90/50	1		\$8,240	\$8,240	East Coast Induction (USA)	—
GEN-2	solid-state RF generator (V3000-0070)						
GEN-2	Power Controller C-V3 Plus (V3000-0414)	1		\$1,670	\$1,670	East Coast Induction (USA)	—
GEN-3	Recirculating water chiller (MIL043008	1		\$1,575	\$1,575	East Coast Induction (USA)	—
GEN-4	air-cooled closed-loop, V1650-0060)						
GEN-4	Heating head + 3 m cable and water lines (PWH-13-12-30/50, V3000-0300); 1 coil made to spec at no charge with a complete unit	1		\$3,337	\$3,337	East Coast Induction (USA)	copper
GEN-5	Line auto-transformer, 12 kVA / 3PH / 480×380 V with taps (V049-0508; facility-dependent)	1		\$1,665	\$1,665	East Coast Induction (USA)	—
RETRO-1	LabVIEW-compatible DAQ with analog out + in (e.g. NI USB-6001/6009-class, 0–5 V AO) [TODO: confirm exact model]	1		[TODO:]	[TODO:]	National Instruments	—
RETRO-2	Dual-wavelength ratio pyrometer (LumaSense IMPAC ISR 6, 800–2500 °C)	1		\$241 (used)	\$241	eBay — used (new list ~\$5,500, LumaSense quote 00161403)	—
RETRO-3	24 V linear power supply for pyrometer (International Power)	1		\$51	\$51	Mouser	—
RETRO-4	Voltage-to-current (0–5 V → 4–20 mA) loop conditioner for the generator power-setpoint input	1		[TODO:]	[TODO:]	[TODO: signal-conditioning vendor]	—
RETRO-5	Edwards T-Station 85 turbo-molecular pump + gauge	1		[TODO:]	[TODO:]	Edwards	—
RETRO-6	Edwards TAV5 vent valve	1		\$59 (used)	\$59	eBay — used (price from purchase record; verify current listing or use a vacuum-supply quote)	—
RETRO-7	Inert-gas regulator (Fisher FS-50)	1		\$38	\$38	Fisher Scientific	—
RETRO-8	KF40 overpressure centering ring	1		\$19	\$19	IdealVac	aluminum
RETRO-9	KF40 plastic quick vacuum clamp	1		\$23	\$23	IdealVac	polymer
RETRO-10	Optical window, custom quartz disc 55 mm × 1.5 mm (1 used; buy 3 for spares)	3		\$18 (est.)	\$54	Custom-cut metric disc — vendor not recorded in parts list; price approximate (McMaster stocks only imperial sizes)	fused quartz
RETRO-11	Ultra-high-temperature quartz disc 2 in × 1/16 in (McMaster 1357T12)	1		\$19	\$19	McMaster-Carr	fused quartz
RETRO-12	Inert-gas plumbing (BSPT/KF adapters, barbs, tee, 10 psi relief valve McMaster 4772K4)	1 set	see parts list		~\$66	McMaster-Carr / BMotionTech	brass / steel
CONS-1	Quartz tube, 4 ft × 35 mm ID, cut to length	2		\$67	\$134	QSI Quartz	fused quartz
CONS-2	Graphite stock for machined crucible/susceptor (56L-3, 3000 °C, 1×6×6 in)	1		\$99	\$99	Cotronics	graphite
CONS-3	Graphite repair cement (Resbond 931-1, 3000 °C)	1		\$108	\$108	Cotronics	graphite
CONS-4	Alumina crucible stock (RTC-60-2, 1787 °C, 10 lb)	1		\$91	\$91	Cotronics	alumina
CONS-5	Zirconia crucible stock (760-1, 2204 °C, 10 lb)	1		\$124	\$124	Cotronics	zirconia
CONS-6	Torr-Seal high-vacuum epoxy / O-rings	1 set		\$57	\$57	Varian / McMaster	epoxy / elastomer

TABLE VIII. Open design files and their locations in the repository.

Design file	File type	License	Location
LabVIEW control VIs (furnace control, manual control, PID tuning v1/v2, email alert)	LabVIEW .vi	MIT (proposed)	code/induction-furnace-control-code/
Manual ramping v5 VIs	LabVIEW .vi	MIT (proposed)	code/manual-ramping-v5/
plotheatcurve.m (heat-curve plotter)	MATLAB	MIT (proposed)	code/plotheatcurve.m
Work-coil drawing	PDF / OXPS	CERN-OHL-S (proposed)	docs/coils-drawing.pdf, docs/coils.oxps
System schematics	PPTX (+ extracted text/figures)	CERN-OHL-S (proposed)	docs/*.pptx, paper/extracted-context/
Temperature-control wiring	PNG	CERN-OHL-S (proposed)	docs/temp-control-modification/0-5V PLC wire design.png
KF40 overpressure centering ring	STEP	CERN-OHL-S (proposed)	docs/KF Supplies/KF40-overpressureCenteringRing.step
Graphite crucible/susceptor machining drawing	[TODO: add]	CERN-OHL-S (proposed)	[TODO:]